RM03, 4, 5

09/10/2024

Code-C_Phase-3

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 NCERT Booster Test Series for NEET (RM)-2024-25_T01C **Time : 200 Min.**

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics
Botany: Cell: The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification, Morphology of Flowering Plants
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules, Breathing & Exchange of Gases, Body Fluids & Circulation

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. If $k\hat{i}+\frac{1}{2}\hat{j}$ is a unit vector, then k can be equal to

(1) $\frac{\sqrt{3}}{2}$

(2) $\frac{1}{2}$

(3) 1

(4) 2

2. A man can swim in still water with a speed 2 m/s. The minimum time, in which he can cross a river of width 200 m is

(1) 100 s

(2) 200 s

(3) 50 s

(4) 75 s

3. A projectile is projected from ground at an angle 45° with horizontal, if range of projectile is R, then the maximum height is

(1) $\frac{R}{4}$

(2) $\frac{R}{2}$

(3) $\frac{R}{8}$

(4) $\frac{R}{6}$

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4. The velocity of A and B are $v_A = \left(2\hat{i} + 3\hat{j}\right)$ m/s and $v_B = \left(3\hat{i} + 4\hat{j}\right)$ m/s. Velocity of B as observed by A is

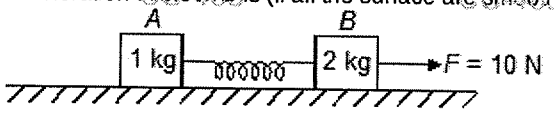
(1) $\left(-\hat{i} + \hat{j}\right)$ m/s

(2) $\left(\hat{i} + \hat{j}\right)$ m/s

(3) $\left(2\hat{i} + 3\hat{j}\right)$ m/s

(4) $\left(-\hat{i} - \hat{j}\right)$ m/s

5. If the acceleration of block A is 2 m/s^2, then the acceleration of block B is (If all the surface are smooth)



(1) 1 m/s^2

(2) 2 m/s^2

(3) 3 m/s^2

(4) 4 m/s^2

6. The momentum of a body in two perpendicular directions at any time t are given as $P_x = 2t^2 + 6$ and $P_y = \frac{3t^2}{2} + 3$ where P_X and P_Y are in kg ms^{-1}. The magnitude of force (in N) acting on the body at $t = 2$ s is

(1) 10

(2) 20

(3) 25

(4) 14

7. In circular motion, which of the following is directed along axis of rotation?

(1) Velocity

(2) Centripetal acceleration

(3) Angular velocity

(4) Tangential acceleration

8. Inside a railway car a bob is suspended from the roof. When the railway car accelerates to the right, then with respect to the railway car

(1) The bob moves vertically downward

(2) The bob moves to the right

(3) The bob moves to the left

(4) The bob remains at rest

9. A stationary bomb explodes into two fragments of masses m_1 and m_2. If momentum of one fragment is p, the total kinetic energy of explosion is

(1) $\frac{p^2(m_1+m_2)}{m_1 m_2}$

(2) $\frac{p^2(m_1+m_2)}{2m_1 m_2}$

(3) $\frac{p^2 m_1 m_2}{m_1+m_2}$

(4) $\frac{2m_1 m_2}{p^2(m_1+m_2)}$

10. Internal force of a system can change

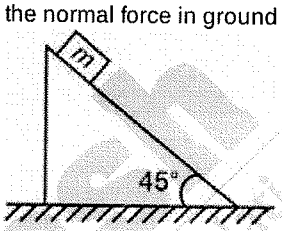
(1) Linear momentum as well as kinetic energy of system

(2) Linear momentum but not the kinetic energy of system

(3) The kinetic energy but not linear momentum of system

(4) Neither the linear momentum nor the kinetic energy of system

11. In the figure shown all the surfaces are frictionless, and mass of the block is m = 5 kg. If block slides along the incline while the wedge is kept fixed, then work done by the normal force in ground frame on the block in $\sqrt{2}$ s is



(1) Zero

(2) 200 J

(3) 100 J

(4) $100\sqrt{3}$ J

12. Which of the following statements is wrong?

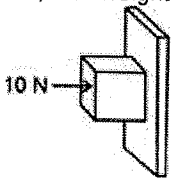
(1) Kinetic energy of a body is independent of the direction of motion

(2) In an elastic collision of two bodies, the momentum and kinetic energy of system is conserved before and after the collision

(3) Kinetic energy can be positive, negative and zero

(4) A body cannot have kinetic energy without momentum

13. A horizontal force of 10 N is necessary to just hold a block stationary against a wall as shown in figure, if the coefficient of friction between the block and the wall is 0.2, then weight of the block is



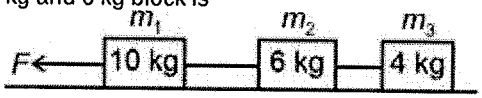
(1) 2 N

(2) 0.2 N

(3) 4 N

(4) 0.1 N

14. Three blocks of masses m_1, m_2 and m_3 are placed on a horizontal frictionless surface. If a force (F) of 40 N pulls the system, then the tension in the string connecting 10 kg and 6 kg block is



(1) 30 N

(2) 20 N

(3) 2 N

(4) 5 N

15. Which among the following is a supplementary unit?

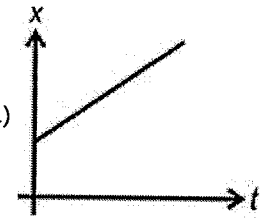
(1) steradian

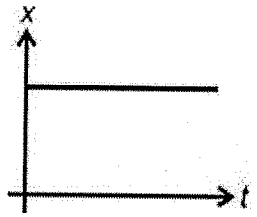
(2) kelvin

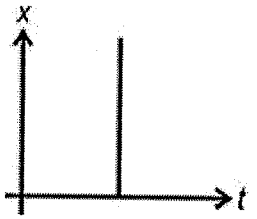
(3) ampere

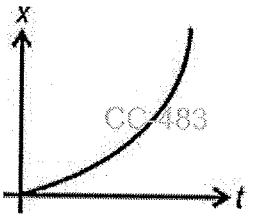
(4) candela

16. Which of the following variations correctly represents the object at rest? [Where symbols have their usual meaning]

(1) 

(2) 

(3) 

(4) 

17. A ball is thrown vertically upward with velocity of 120 m s^{-1}. If retardation due to air resistance is 2 m/s^2, then total time taken by ball to reach maximum height is

(1) 25 s

(2) 10 s

(3) 20 s

(4) 15 s

18. A scooter accelerates from rest for time t_1 at constant rate a_1 and then retards at constant rate a_2 for time t_2 and eventually comes to rest. The correct value of $\frac{a_1}{a_2}$ will be

(1) $\frac{t_1+t_2}{t_1}$

(2) $\frac{t_1}{t_2}$

(3) $\frac{t_2}{t_1}$

(4) $\frac{t_1+t_2}{t_2}$

19. Position of a particle moving along a straight line is given by $x = (40 + 27t - t^3)$ m where t is in seconds. The position of particle when it comes to rest is

(1) 27 m

(2) 30 m

(3) 94 m

(4) 54 m

20. If in a vernier calliper 10 VSD coincides with 8 MSD, then least count of vernier calliper is [1 MSD = 1 mm]

(1) 0.01 mm

(2) 0.2 mm

(3) 0.3 mm

(4) 0.8 mm

21. The length and breadth of a rectangular plate are measured to be 14.5 cm and 4.2 cm respectively. The area of the plate to appropriate significant figure is

(1) 60.9 cm^2

(2) 60 cm^2

(3) 61 cm^2

(4) 60.98 cm^2

22. In an experiment, four quantities a, b, c and d are measured with percentage errors of 2%, 3%, 1% and 0.5% respectively. A quantity X is defined as $X = \frac{a\sqrt{b}}{c^{\frac{3}{2}}d^4}$.

The maximum percentage error in the measurement of X will be

(1) 5%

(2) 6%

(3) 7%

(4) 9%

23. The number of significant zeros present in the measured value 0.020040, is

(1) Five

(2) Two

(3) One

(4) Three

24. A person running due east at a speed 6 m/s feels the wind is blowing towards south at a speed 6 m/s. Find the actual velocity of wind blow.

(1) $6\sqrt{2}$ m/sin S-E direction

(2) $3\sqrt{2}$ m/s in S-E direction

(3) 4 m/s in N-E direction

(4) 5 m/s in N-E direction

25. If energy [*E*], velocity [*V*] and time [*T*] are chosen as fundamental quantities, then dimensional formula of force is

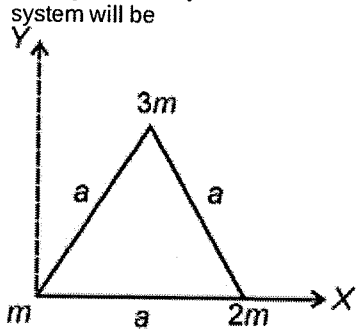
(1) $[EV^{-1}T^{-1}]$

(2) $[EV^{-1}T]$

(3) $[EV^{-1}T^{-2}]$

(4) $[EVT^{-2}]$

26. Three objects having mass *m*, 2*m* and 3*m* are placed at the corner of an equilateral triangle of side *a* as shown in the figure. The y-coordinate of centre of mass of the system will be



(1) $\frac{a}{2}$

(2) $\frac{\sqrt{3}a}{4}$

(3) $\frac{a}{\sqrt{3}}$

(4) Zero

27. Two point masses 2 kg and 3 kg at an instant are located at –2 m and 2 m on *x*-axis respectively. If both are moving in –ve *y*-axis direction with 2.5 m/s and 2 m/s velocities respectively. The angular momentum of the system about origin. (In kg m^2/s) is

(1) $6\hat{k}$

(2) $-2\hat{k}$

(3) $4\hat{k}$

(4) $-4\hat{k}$

28. The centre of mass of a body

(1) Always lie at the geometric centre

(2) Always lie inside the body

(3) Always lie outside the body

(4) May lie outside the body

29. Forces which obey inverse square law are

(1) Gravitational force

(2) Electromagnetic force

(3) Nuclear force

(4) Both (1) and (2)

30. The magnitude of cross product of two parallel vectors is equal to

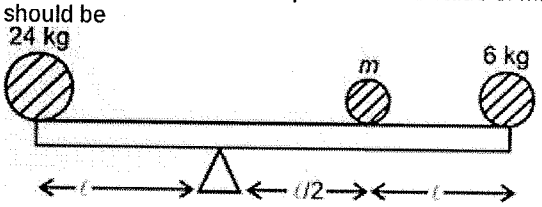
(1) One

(2) Zero

(3) Product of their magnitudes

(4) Sum of their magnitudes

31. Three masses are placed on light rod as shown. For the system to be in rotational equilibrium the value of mass *m* should be



(1) 30 kg

(2) 15 kg

(3) 21 kg

(4) 1 kg

32. Wheel of an engine rotates with an angular speed of 120 rev/min. If radius of the wheel is 2 m, then linear speed of any point on its rim is

(1) 16π m s^{-1}

(2) 8π m s^{-1}

(3) 4π m s^{-1}

(4) 2π m s^{-1}

33. Angular speed of a rotating wheel changes from 1500 rpm to 2400 rpm in 6 s. Average angular acceleration of rotating wheel is

(1) 4π rad s^{-2}

(2) 6π rad s^{-2}

(3) 5π rad s^{-2}

(4) 10π rad s^{-2}

34. A body of moment of inertia 2 kg m^2 is rotated at a speed of 25 rad s^{-1}. A tangential force at rim stops the wheel in 10 s. Average torque of force is

(1) 5 N m

(2) 10 N m

(3) 15 N m

(4) 2.5 N m

35. The potential energy of a 1 kg particle, free to move along x-axis is given by $U(x) = \left(\frac{x^3}{3} - \frac{x^2}{2}\right)$ J.

Among the following options, the particle will be in equilibrium at (x is in meter)

(1) $x = 1$ m

(2) $x = -1$ m

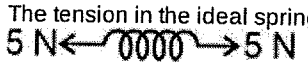
(3) $x = 2$ m

(4) $x = -2$ m

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SECTION-B

36. The tension in the ideal spring as shown in the figure is



(1) Zero

(2) 2.5 N

(3) 5 N

(4) 10 N

37. Consider the following statements and choose the correct option.

(I) Non-zero net force acting on a particle must change its linear momentum.

(II) An external force acting on a particle certainly changes its kinetic energy.

(1) Only (I) is correct

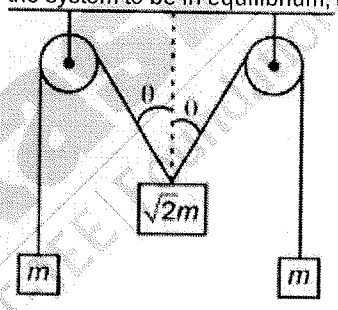
(2) Only (II) is correct

(3) Both (I) and (II) are correct

(4) Both (I) and (II) are incorrect

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38. Three blocks having masses m, $\sqrt{2}m$ and m are connected by a light string which is passing through two massless and frictionless pulleys as shown in figure. For the system to be in equilibrium, the angle θ should be



(1) 15°

(2) 30°

(3) 45°

(4) 60°

39. Time intervals measured by a clock gives the following readings:

1.25 s, 1.24 s, 1.27 s, 1.21 s and 1.28 s. What is the absolute mean error in the observations?

(1) 0.02

(2) 0.05

(3) 0.06

(4) 0.03

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40. If the equation of motion of projectile in *x-y* plane is given by $y = 2x - 3x^2$, then the angle of projection with *x*-axis is

(1) $\tan^{-1}$ (3)

(2) $\tan^{-1}$ (2)

(3) $\sin^{-1}$ (2)

(4) $\cos^{-1}$ (2)

41. A car moves along a curved road of radius 100 m with 20 m/s. It is increasing its speed at the rate of 3 m/s^2. What is the magnitude of its net acceleration?

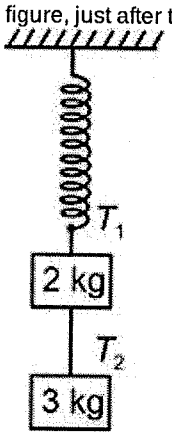
(1) Zero

(2) 2 m/s^2

(3) 5 m/s^2

(4) 8 m/s^2

42. Find acceleration of block of mass 2 kg shown in the figure, just after the string is cut.



(1) 7 m/s^2 downwards

(2) 15 m/s^2 upwards

(3) 10 m/s^2 upwards

(4) 25 m/s^2 downwards

43. Speed of boat in still water is 2 km/h and width of river is 500 m. If it takes 30 minutes to cross the river along the shortest path, then the velocity of the river is

(1) $\sqrt{2}$km/h

(2) $\sqrt{3}$km/h

(3) 1 km/h

(4) $\sqrt{5}$km/h

44. A screw gauge has least count of 0.05 mm and there are 100 divisions on its circular scale. The pitch of screw gauge is

(1) 0.25 mm

(2) 0.30 mm

(3) 5 mm

(4) 0.05 mm

45. A particle is moving on a straight line and has initial velocity equal to 5 m/s. If the particle has constant acceleration equal to 4 m/s^2, then the distance covered by it in 3rd second is

(1) 10 m

(2) 15 m

(3) 7 m

(4) 12 m

46. The moment of inertia of uniform rectangular plate having length ℓ and width *b* about an axis passing through its centre and parallel to its length ℓ is (*M* is mass of plate)

(1) $\frac{M\ell^2}{12}$

(2) $\frac{Mb^2}{12}$

(3) $\frac{M\ell^2}{6}$

(4) $\frac{Mb^2}{6}$

47. A sphere is moving towards (+ve) *x*-axis with a velocity v_C and rotates anticlockwise with angular speed 'ω' such that $v_C = \frac{\omega R}{3}$. Speed of bottom point of sphere is

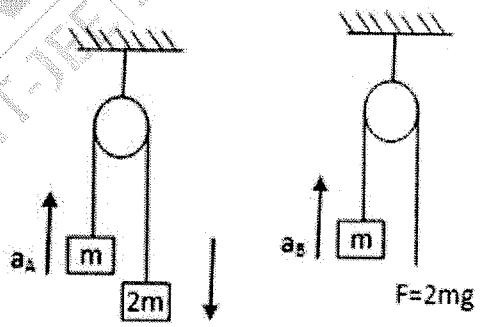
(1) ωR

(2) $\frac{2\omega R}{3}$

(3) $\frac{4\omega R}{3}$

(4) $\frac{\omega R}{3}$

48. Two diagrams as shown in the figure, find the relation between a_A&a_B



(i) (ii)

(1) $a_A = a_B$

(2) $a_A = \frac{a_B}{2}$

(3) $a_A = \frac{a_B}{3}$

(4) $a_A = 2a_B$

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49. A bullet after entering a wooden block covers 15 cm inside it and loses half of its velocity. After how much further distance will it come to stop?

(1) 3 m

(2) 5 cm

(3) 7 cm

(4) 10 cm

50. Two bodies of masses 4 kg and 5 kg are kept at fixed locations having position vectors $-\hat{i}+\hat{j}+2\hat{k}$ and $4\hat{i}-3\hat{j}+\hat{k}$ respectively. The position vector of centre of mass of this system is

(1) $\frac{1}{9}[16\hat{i}+11\hat{j}+13\hat{k}]$

(2) $\frac{1}{9}[16\hat{i}-11\hat{j}+13\hat{k}]$

(3) $\frac{1}{9}[11\hat{i}+16\hat{j}+32\hat{k}]$

(4) $\frac{1}{9}[16\hat{j}+11\hat{k}]$

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CHEMISTRY

SECTION-A

51. Number of electrons ejected from a metal surface during photoelectric effect depends on

(1) Intensity of incident light

(2) Wavelength of incident light

(3) Frequency of incident light

(4) Energy of incident light

52. The correct set of four quantum numbers for the valence electron of potassium atom ($Z = 19$) is

(1) $4, 1, 0, +\frac{1}{2}$

(2) $4, 0, 0, +\frac{1}{2}$

(3) $4, 0, 1, +\frac{1}{2}$

(4) $3, 1, 0, -\frac{1}{2}$

53. The radii of 2nd Bohr orbit of Be^{3+} ion is

(1) 26.45 pm

(2) 52.9 pm

(3) 79.35 pm

(4) 105.8 pm

54. Number of angular and radial nodes present in 5*p* orbital are respectively

(1) 2 and 1

(2) 1 and 2

(3) 1 and 3

(4) 3 and 1

55. Which among the following electromagnetic radiations has minimum wavelength?

(1) Radio waves

(2) γ - rays

(3) Microwaves

(4) UV rays

56. Spectral line of which series does not belong to infrared region?

(1) Pfund

(2) Brackett

(3) Paschen

(4) Balmer

57. A golf ball has a mass of 50 g, and a speed of 30 m/s. If the speed can be measured within accuracy of 1%, then the uncertainty in the position will be

(1) 3.5×10^{-33} m

(2) 7.5×10^{-33} cm

(3) 2×10^{-33} m

(4) 1.39×10^{33} km

58. Empirical formula of the compound is CH_2O. Which of the following molecular formula is not possible?

(1) $C_6H_{12}O_6$

(2) HCHO

(3) CH_3COOH

(4) $C_{12}H_{22}O_{11}$

59. 100 ml of 2 M H_2SO_4 solution is diluted by adding 400 ml water. The molarity of the resultant solution becomes

(1) 0.4 M

(2) 0.6 M

(3) 0.75 M

(4) 0.2 M

60. An element X has two isotopes ^{52}X and ^{54}X. If percentage abundance of lighter isotope is 60% then the average atomic mass of the element X will be

(1) 53.4

(2) 52.4

(3) 52.8

(4) 53.2

61. 23 g of NO_y occupies 11.2 litres at STP. Assuming ideal gas nature, the value of *y* is

(1) 1

(2) 2

(3) 3

(4) 4

62. Which of the following is an incorrect statement about atom as per Daltons Atomic Theory?

(1) Matter consists of indivisible atoms

(2) Atoms of different elements have identical mass

(3) Compound are formed when atoms of different elements combine in a fixed ratio

(4) Atoms are neither created nor destroyed in a chemical reaction

63. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): 0.031 has four significant figures.
Reason (R): Zeros preceding to first non-zero digit are significant.
In the light of above statements choose the correct answer.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

64. In the following questions, a statement of assertion (A) is followed by a statement of reason (R).
Assertion (A) : The second Ionisation enthalphy of an element will be lesser than the first Ionisation enthalphy.
Reason (R) : It is more easier to remove an electron from a positively charged ion than from the neutral atom.
In the light of above statements, choose the correct answer.

(1) Both (A) and (R) are true and the (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true but the (R) is not the correct explanation of the (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

65. Identify the incorrect match from the options.

	Name		IUPAC Official Name
a.	Unnilbium	(i)	Nobelium
b.	Unnilquadium	(ii)	Rutherfordium
c.	Unnilseptium	(iii)	Bohrium
d.	Unnilennium	(iv)	Dubnium

(1) a, (i)

(2) b, (ii)

(3) c, (iii)

(4) d, (iv)

66. Which pair contains representative elements?

(1) Sc and Ti

(2) Mg and S

(3) P and Fe

(4) Cd and Hg

67. Which of the following has a positive value of electron gain enthalpy?

(1) Hydrogen

(2) Sulphur

(3) Neon

(4) Oxygen

68. Correct decreasing order of ionic radii is

(1) $Na^+ > Mg^{2+} > F^- > O^{2-}$

(2) $O^{2-} > F^- > Na^+ > Mg^{2+}$

(3) $Mg^{2+} > Na^+ > F^- > O^{2-}$

(4) $O^{2-} > F^- > Mg^{2+} > Na^+$

69. Addition of an electron to a neutral gaseous atom always leads to
(I) Formation of an anion.
(II) Liberation of energy.
(III) Decrease in proton/electron ratio.

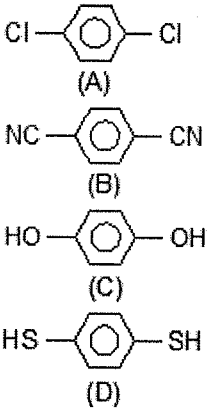
(1) Only (II)

(2) (I) & (II)

(3) (I) & (III)

(4) Only (I)

70. For which of the following molecules μ(Dipole moment) is/are non-zero?



(1) Only (A)

(2) Both (A) & (B)

(3) Only (C)

(4) Both (C) & (D)

71. Paramagnetic species among the following is

(1) O_2

(2) C_2

(3) C_2^{2-}

(4) O_2^{2-}

72. The correct order of bond angle in CH_4, NH_3, H_2O and BF_3 is

(1) $CH_4 > NH_3 > H_2O > BF_3$

(2) $BF_3 > CH_4 > NH_3 > H_2O$

(3) $H_2O > NH_3 > CH_4 > BF_3$

(4) $NH_3 > CH_4 > BF_3 > H_2O$

73. Given below are two statements one is lablled as assertion (A) other is reason (R)
Assertion (A): Dipole moment of NH_3 is greater than NF_3
Reason (R): NH_3 is pyramidal but NF_3 is planar molecule
In the light of above statements choose the correct answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

74. Which of the following molecules contains equal number of σ and π-bonds?

(1) XeO_2F_2

(2) $POCl_3$

(3) $SOCl_2$

(4) SO_3

75. Consider the following statements
Statement I: Total number of electrons present in π antibonding molecular orbital of N_2 is zero.
Statement II: Total number of antibonding electrons in N_2 is 4.
In light of above statements mark the correct statements.

(1) Both the statement I and II are correct

(2) Both the statement I and II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

76. The bond order of P – O bond in PO_4^{3-} is

(1) 1.5

(2) 2

(3) 1.25

(4) 1.75

77. Under what conditions, a reaction is spontaneous at all temperatures?

(1) $\Delta H > 0, \Delta S > 0$

(2) $\Delta H < 0, \Delta S < 0$

(3) $\Delta H > 0, \Delta S < 0$

(4) $\Delta H < 0, \Delta S > 0$

78. From the following bond energies
H – H : 430 kJ/mol
C = C : 600 kJ/mol
C – C : 330 kJ/mol
C – H : 410 kJ/mol
Enthalpy for the reaction
$H_2C = CH_2 + H_2 \rightarrow H_3C - CH_3$ will be

(1) –120 kJ/mol

(2) –240 kJ/mol

(3) –100 kJ/mol

(4) –360 kJ/mol

79. Given below are two statements: One is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): A water filled bucket represents an open system.
Reason (R): In an open system only energy is exchanged between system and surroundings.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

80. An ideal gas is allowed to expand isothermally against a constant external pressure of 2.5 atm from an initial volume of 2.5 L to final volume of 4.5 L. ΔU for this process is

(1) – 505 J

(2) 505 J

(3) 113.625 J

(4) 0 J

81. If $\Delta_f G°$ for any reaction is –2.303 J then, the value of equilibrium constant K is

(1) 10^{RT}

(2) $10^{2.303/RT}$

(3) $10^{1/RT}$

(4) $10^{RT/2.303}$

82. If enthalpy of hydrogenation of cyclohexene is –x kJ mol^{-1} and resonance energy of benzene is –y kJ mol^{-1} then the enthalpy of hydrogenation of benzene will be

(1) (y – 3x) kJ mol^{-1}

(2) –(x + y) kJ mol^{-1}

(3) (y + 3x) kJ mol^{-1}

(4) (x + y) kJ mol^{-1}

83. For which of the following reactions, ΔH is not equal to ΔE?

(1) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

(2) $H_2(g) + I_2(g) \rightarrow 2\,HI(g)$

(3) $N_2(g) + 3H_2(g) \rightarrow 2\,NH_3(g)$

(4) $C(s) + O_2(g) \rightarrow CO_2(g)$

84. Correct order of electronegativity of the given elements is

(1) P > Si > C

(2) C > P > Si

(3) P > C > Si

(4) C > Si > P

85. Given below are two statements one is labelled as assertion (A) and other is labelled as reason (R).
Assertion (A): On changing the concentration of limiting reagent, amount of product will also be changed.
Reason (R): Limiting reagent determines yield of products.
In the light of above statements choose the correct answer.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

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SECTION-B

86. Maximum number of electrons associated with $\ell = 1$, m = 0 in magnesium atom is

(1) 1

(2) 2

(3) 6

(4) 8

87. Match List-I with List-II and choose the correct code.

	List-I (Orbitals)		List-II (Number of radial nodes)
(a)	2*p*	(i)	1
(b)	3*p*	(ii)	3
(c)	3*s*	(iii)	0
(d)	4*s*	(iv)	2

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

88. A 200 W bulb emits monochromatic light of wavelength 1400 Å and only 10% of the energy is emitted as light. The number of photons emitted by the bulb per second will be

(1) 1.4×10^{18}

(2) 1.4×10^{20}

(3) 1.4×10^{19}

(4) 1.4×10^{21}

89. Assuming the density of H_2O is equal to 1 g/mL, the total number of atoms present in 1 litre H_2O is

(1) 166.66 N_A

(2) 55.5 N_A

(3) 111 N_A

(4) 18.5 N_A

90. Chlorophyll is a biomolecule responsible for green colour in the plants. Chlorophyll contains 2.68% of magnesium by mass. Calculate the number of magnesium atoms in 6 g of chlorophyll (Mg = 24)

(1) 2.01×10^{21}

(2) 4.03×10^{21}

(3) 6.02×10^{23}

(4) 4.03×10^{23}

91. Identify the incorrect match.

	Multiple	Prefix
(1)	10^{-18}	atto
(2)	10^{-15}	pico
(3)	10^{-21}	zepto
(4)	10^{-9}	nano

(1) (1)

(2) (2)

(3) (3)

(4) (4)

92. Which of the following oxides is most basic?

(1) P_4O_{10}

(2) N_2O_5

(3) CO_2

(4) CaO

93. Electronic configuration of elements A, B, C and D are as follows

A : $1s^2 2s^2 2p^4$

B : $1s^2 2s^2 2p^5$

C : $1s^2 2s^2 2p^6 3s^2 3p^5$

D : $1s^2 2s^2 2p^3$

The most electronegative element among the following is

(1) A

(2) B

(3) C

(4) D

94. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be

(1) –5.1 eV

(2) –10.2 eV

(3) +2.55 eV

(4) –2.55 eV

95. Which of the following molecular orbitals contain zero nodal plane?

(1) σ^*1s

(2) $\sigma 2s$

(3) $\pi 2p_x$

(4) $\pi^* 2p_x$

96. Given below are two statements: one is labelled as assertion (A) and the other is labelled as reason (R).

Assertion (A): In PCl_5 molecule, axial bonds are weaker than the equatorial bonds.

Reason (R): In trigonal bipyramidal geometry, the axial bond pairs suffer more repulsive interaction from the equatorial bond pairs.

In the light of above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

97. The number of 90° and 180° angles found in sp^3d^2 hybridisation respectively are

(1) 6, 3

(2) 8, 4

(3) 8, 3

(4) 12, 3

98. Given below are two statements.
Statement-I: T, P and V are called state variables or state functions.
Statement-II: Values of state functions depend on the state of the system and how it is reached.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

99. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): During adiabatic compression, heating takes place.
Reason (R): Work done on the system during adiabatic process increases internal energy of the system.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true but R is NOT the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A

100. Match the following and choose the correct option.

Reaction	Enthalpy
A. $Cl_2(g) \rightarrow 2\,Cl(g)$	(i) $\Delta_{eg}H^\circ$
B. $Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$	(ii) Δ_aH°
C. $Ca(s) + C(graphite) + \frac{3}{2}O_2(g) \rightarrow CaCO_3(s)$	(iii) $\Delta_{lattice}H^\circ$
D. $Cl(g) \rightarrow Cl^-(g)$	(iv) Δ_fH°

(1) A(iii), B(ii), C(iv), D(i)

(2) A(ii), B(iii), C(iv), D(i)

(3) A(ii), B(iv), C(i), D(iii)

(4) A(iv), B(iii), C(ii), D(i)

BOTANY

SECTION-A

101. Microbodies

(1) Are membraneless

(2) Contain various enzymes

(3) Are present only in animal cell

(4) Are site for protein synthesis in prokaryotes

102. Match the following columns and select the **correct** option.

Column I	Column II
a. Chloroplast	(i) Cristae
b. Plasmodesmata	(ii) Thylakoid
c. Mitochondria	(iii) Calcium pectate
d. Middle lamella	(iv) Connects cytoplasm of adjacent cells

(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iv), b(i), c(ii), d(iii)

103. Oil storing leucoplasts are called

(1) Elaioplasts

(2) Amyloplasts

(3) Chromoplasts

(4) Aleuroplasts

104. Select the **incorrect** statement w.r.t. plasma membrane.

(1) Proteins embedded in membrane are intrinsic proteins

(2) Phospholipid head is polar in nature

(3) Phospholipid tails are hydrophilic

(4) Proteins present on the surface are peripheral proteins

105. Read the below given statements and select the **correct** option.
Statement A : The primary cell wall of a plant cell is capable of growth.
Statement B : The middle lamella glues the different neighbouring cells together.

(1) Only A is incorrect

(2) Only B is incorrect

(3) Both A & B are correct

(4) Both A & B are incorrect

106. Select the **mismatched** pair.

(1) Vacuole – Tonoplast

(2) Inclusion body – Found only in eukaryotic cells

(3) Chromatophores – Blue-green algae

(4) Mitochondria – Sausage-shaped

107. Cell theory was given by

(1) Matthias Schleiden and Robert Hooke

(2) Robert Hooke and Theodore Schwann

(3) Theodore Schwann and Leeuwenhoek

(4) Matthias Schleiden and Theodore Schwann

108. The smallest living cell

(1) Is a prokaryote

(2) Has mitochondria but no chloroplast

(3) Lacks cell membrane

(4) Lacks genetic material

109. Endomembrane system of cell includes all, **except**

(1) Golgi bodies

(2) ER

(3) Mitochondria

(4) Lysosomes

110. Splitting of centromeres and separation of chromatids occurs at

(1) Telophase

(2) Anaphase

(3) Prophase

(4) Metaphase

111. Dyad of cells appears at

(1) Telophase I

(2) Telophase II

(3) Prophase I

(4) Anaphase I

112. Chromosomes are attached to the spindle fibres by

(1) Chromatids

(2) Kinetochores

(3) Satellite

(4) Telomere

113. In final stage of prophase-I of meiosis

(1) Chiasmata formation occurs

(2) Crossing over occurs

(3) Recombinase enzyme gets fully activated

(4) Terminalisation of chiasmata takes place

114. When karyokinesis is **not** followed by cytokinesis, it results in the

(1) Entrance of the cell into G_0 phase

(2) Formation of cell plate

(3) Haploid stage of the cell

(4) Formation of syncytium

115. All the following events occur in the resting phase of cell cycle, **except**

(1) Condensation of chromatin material

(2) Duplication of cell organelles

(3) Duplication of genetic material

(4) Growth of cell in size

116. Select the option showing hierarchial arrangement of taxonomic categories in ascending order

(1) Species → Genus → Family → Order

(2) Genus → Species → Family → Order

(3) Genus → Species → Order → Family

(4) Species → Family → Order → Genus

117. ICBN stands for

(1) International Code for Bacterial Nomenclature

(2) Indian Code for Botanical Nomenclature

(3) International Code for Biological Nomenclature

(4) International Code for Botanical Nomenclature

118. Correct representation of scientific name of wheat is

(1) *Triticum Aestivum*

(2) Triticum aestivum

(3) *triticum aestivum*

(4) *Triticum aestivum*

119. Sapindales and Poales belong to

(1) Same class and division

(2) Different classes and divisions

(3) Same class but different divisions

(4) Different classes but same division

120. Read the following statements and select the **correct** option.
Statement A : Bacteria as a group show most extensive metabolic diversity.
Statement B : Bacteria have simple behaviour but very complex structure.

(1) Only A is correct

(2) Only B is correct

(3) Both A and B are correct

(4) Both A and B are incorrect

121. Mycelium is aseptate and coenocytic in

(1) *Penicillium*

(2) *Albugo*

(3) *Agaricus*

(4) *Aspergillus*

122. *Nostoc* lacks

(1) Heterocysts

(2) Mucilagenous sheath

(3) Photosynthetic pigments

(4) Heterotrophic mode of nutrition

123. Which of the following is **not** true for deuteromycetes?

(1) Asexually reproduce by conidia

(2) Mycelium is septate and unbranched

(3) Some members are saprophytic or parasite while a large number of them are decomposers of litter

(4) Commonly known as imperfect fungi

124. The cell wall resembles a soap box in

(1) Diatoms

(2) Dinoflagellates

(3) Slime moulds

(4) *Gonyaulax*

125. Which of the following is **not** the basis of classification of fungi?

(1) Morphology of the mycelium

(2) Mode of spore formation

(3) Mode of nutrition

(4) Fruiting bodies

126. The function of ribosomes of a polysome is to

(1) Translate the mRNA into protein

(2) Synthesise lipids in eukaryotic cells

(3) Replicate DNA during cell division

(4) Store various hydrolytic enzymes

127. Tonoplast

(1) Is the membrane of gas vacuoles.

(2) Helps in excretion in *Amoeba*

(3) Facilitates transports of ions against the concentration gradient

(4) Is absent in plant cell vacuole

128. What will be the amount of DNA in each meiosis II product if the mother cell has 40 pg of DNA in G_2 phase?

(1) 10 pg

(2) 20 pg

(3) 40 pg

(4) 80 pg

129. Read the following statements and choose the **correct** option.
Assertion : Root cap protects the tender apex of the root.
Reason : Root hair increases the surface area for absorbtion of water and minerals from the soil.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

130. The edible part of mango fruit is

(1) Endosperm

(2) Endocarp

(3) Mesocarp

(4) Epicarp

131. Perigynous condition is seen in

(1) Guava

(2) Cucumber

(3) Plum

(4) Brinjal

132. In Solanaceae family flowers have

(1) Epitepalous stamens

(2) Actinomorphic symmetry

(3) Unisexual flower

(4) Monocarpellary ovary

133. In roots, the region of meristematic activity

(1) Has cells which do not divide at all

(2) Has larger cells with thick wall and dense cytoplasm

(3) Is present few millimetres above the root cap

(4) Is directly responsible for the formation of root hairs

134. 'Margins of petals overlap each other, but not in a fix pattern' is true for

(1) China rose

(2) Mustard

(3) *Cassia*

(4) Bean

135. Arrangement of veins and veinlets in leaf lamina is called

(1) Venation

(2) Phyllotaxy

(3) Aestivation

(4) Placentation

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SECTION-B

136. The core of eukaryotic cilia and flagella is called axoneme which is made up of

(1) Filament, hook and basal body

(2) 9 centrally located doublets and 2 microtubules at periphery

(3) 2 centrally located microtubules and 9 doublets at periphery

(4) Central hub and 11 peripheral doublets

137. The type of chromosome in which the centromere divides the chromosome into two equal arms is called

(1) Acrocentric chromosome

(2) Sub-metacentric chromosome

(3) Metacentric chromosome

(4) Telocentric chromosome

138. An onion root cell has 16-chromosomes in its G_1 phase then how many chromosomes it has in S and G_2 phase respectively?

(1) 16 and 16

(2) 32 and 32

(3) 8 and 8

(4) 32 and 16

139. Find out the **correct** sequence of events of meiosis – I.

A. Condensation of chromatin material

B. Separation of non-sister chromatids of homologous chromosomes towards the opposite poles

C. Degeneration of nuclear membrane

D. Crossing over

(1) A → B → C → D

(2) A → D → C → B

(3) A → D → B → C

(4) C → A → B → D

140. Most of the cell organelle's duplication occurs in

(1) G_2 phase

(2) G_1 phase

(3) S phase

(4) M phase

141. Match the following columns and select the **correct** option.

	Column-I		Column-II
(a)	Family	(i)	Primata
(b)	Order	(ii)	*Solanum*
(c)	Genus	(iii)	Insecta
(d)	Class	(iv)	Anacardiaceae

(1) a(iv), b(i), c(ii), d(iii)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(i), b(ii), c(iii), d(iv)

142. Systematics differs from taxonomy as the former includes

(1) Identification of organisms

(2) Characterisation of organisms

(3) Nomenclature of organisms

(4) Evolutionary relationship among organisms

143. No virus contains both

(1) RNA and protein coat

(2) DNA and protein coat

(3) RNA and DNA

(4) Capsid and double stranded DNA

144. Which of the following organisms are present in the gut of several ruminant animals and also used for biogas production?

(1) Halophiles

(2) Thermoacidophiles

(3) Eubacteria

(4) Methanogens

145. Read the following statements and select the **correct** option.
Assertion (A) : Viruses are inert outside their specific host cell.
Reason (R) : Viruses are facultative parasites.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

146. Read the given statements and choose the **correct** option.
Statement A: Protozoans are heterotrophs.
Statement B: Dinoflagellates release toxin into the sea water which is highly poisonous to some vertebrates.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

147. Sleeping sickness is caused by a

(1) Sporozoan

(2) Ciliated protozoan

(3) Flagellated protozoan

(4) Dinoflagellates

148. Whorled phyllotaxy is seen in

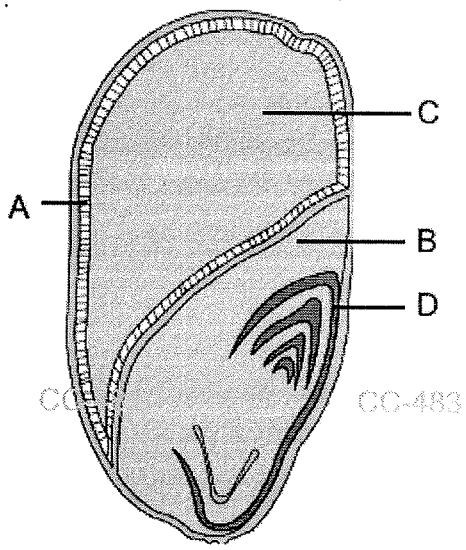
(1) *Calotropis*

(2) Guava

(3) *Alstonia*

(4) China rose

149. Consider the given diagram and select the **correct** option.



(1) A - Lipid layer, separates the embryo

(2) B - Shield shaped cotyledon

(3) C - Bulky diploid cotyledon

(4) D - Radicle enclosing sheath

150. In cymose inflorescence
(a) Main axis terminates into a flower.
(b) Flowers are borne in basipetal order.
(c) Younger flowers are present towards the apex and older at the base.

(1) Only (c) is correct

(2) All (a), (b) and (c) are correct

(3) Only (a) and (b) are correct

(4) Only (a) is correct

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ZOOLOGY

SECTION-A

151. Tendons attach

(1) Skeletal muscles to bones

(2) Smooth muscles to skeletal muscles

(3) Bones to bones

(4) Skeletal muscles to skin

152. Matrix or ground substance of connective tissue is primarily formed by accumulation of

(1) Blood

(2) Modified complex lipids

(3) Modified polysaccharides

(4) Adipocytes

153. Select the **odd one** w.r.t. simple epithelium.

(1) Walls of blood vessels

(2) Inner surface of bronchioles

(3) Moist surface of buccal cavity

(4) Air sacs of lungs

154. A histologist observes epithelium containing single layer of tall and slender cells, having basal nuclei and mucous secreting goblet cells in between them on a slide.
Epithelium being discussed in above statement is

(1) Simple squamous epithelium

(2) Simple cuboidal epithelium

(3) Stratified squamous epithelium

(4) Columnar glandular epithelium

155. The tissue which has a free surface, facing either a body fluid or the outside environment is

(1) Muscular tissue

(2) Connective tissue

(3) Epithelial tissue

(4) Neural tissue

156. Read the statements (A) and (B) w.r.t exothermic reactions and select the **correct** option
(A) Energy content of substrate is lower than that of the product.
(B) External supply of energy is not required to form the product.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct but B is incorrect

(4) Statement A is incorrect but B is correct

157. Nucleosides upon hydrolysis does **not** yield

(1) Pentose sugar

(2) Purine

(3) Pyrimidine

(4) Phosphoric acid

158. Biomolecule that is used for making paper from plant pulp is

(1) Inulin

(2) Cellulose

(3) Glycogen

(4) Chitin

159. Consider the following statements w.r.t. secondary metabolites.
a. Concanavalin A is a lectin.
b. Some secondary metabolites have ecological importance.
c. Cellulose is an alkaloid.
d. Codeine is a pigment.
Choose the option which includes all **correct** statements.

(1) Only a and b

(2) Only a, b and d

(3) Only b, c and d

(4) a, b, c and d

160. Select the biomolecule **not** formed by repetition of monomeric unit *i.e.*, glucose

(1) Starch

(2) Inulin

(3) Glycogen

(4) Cellulose

161. How many water molecules are liberated out in polymerization of 10 monosaccharide units?

(1) 9

(2) 10

(3) 4

(4) 8

162. In a polysaccharide chain such as glycogen, the right end is called the _____ and the left end is called the _____ respectively.

(1) Reducing end, reducing end

(2) Non-reducing end, non-reducing end

(3) Non-reducing end, reducing end

(4) Reducing end, non-reducing end

163. Which of the following is a pentose sugar?

(1) Fructose

(2) Ribose

(3) Glucose

(4) Sucrose

164. Enzyme along with its co-factor is known as

(1) Apoenzyme

(2) Holoenzyme

(3) Prosthetic group

(4) Co-enzyme

165. Choose the **correct** option to complete the analogy.
Nitrogenous base : Adenine :: Nucleoside : _____

(1) Adenosine

(2) Adenylic acid

(3) Deoxyadenylic acid

(4) Guanine

166. In case of serine, R-group of amino acid is

(1) $-CH_3$

(2) $-H$

(3) $-SH$

(4) $-CH_2OH$

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167. The value of tidal volume (TV) in a normal adult male is

(1) 1000 mL

(2) 2000 mL

(3) 500 mL

(4) 1500 mL

168. Every 2 L of oxygenated blood can deliver about 'X' mL of oxygen to the tissues under normal conditions in humans. Choose the option which represents **correct** value for 'X'.

(1) 80

(2) 100

(3) 12

(4) 120

169. Ribs and sternum are lifted upwards due to contraction of

(1) Diaphragm

(2) External inter-costal muscles

(3) Abdominal muscles

(4) Internal inter-costal muscles

170. **Assertion :** Respiratory rhythm is primarily maintained by respiratory rhythm centre present in medulla of the brain.
Reason : Pneumotaxic centre in pons region of the brain can moderate the functions of respiratory rhythm centre.
In the light of above statements, select the **correct** option.

(1) Both Assertion and Reason are false statements

(2) Assertion is true statement but Reason is false

(3) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion

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(4) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion

171. The numerical value of which of the following pulmonary capacities is least in a healthy human at rest?

(1) Expiratory capacity

(2) Functional residual capacity

(3) Inspiratory capacity

(4) Vital capacity

172. The factor that shifts oxygen–haemoglobin dissociation curve to the left side is

(1) High partial pressure of oxygen

(2) High partial pressure of carbon dioxide

(3) High body temperature

(4) High H^+ concentration

173. Choose the most appropriate option to complete the analogy w.r.t. respiratory disorders.
Constricted bronchioles : Asthma : : Damaged alveolar walls: _____

(1) Silicosis

(2) Pleurisy

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(3) Emphysema

(4) Asbestosis

174. How much percentage of O_2 is transported through plasma in dissolved form ?

(1) 97%

(2) 3%

(3) 20-25%

(4) 7%

175. In Coronary Artery Disease

(1) Blood supply to the heart muscle is increased

(2) Lumen of coronary arteries become wider

(3) There is removal of calcium, fat, cholesterol and fibrous tissue in urine

(4) Blood supply to the heart muscle is reduced

176. Neural signals through the sympathetic nerve fibres can increase all of the following, **except**

(1) Heart rate

(2) Duration of cardiac cycle

(3) Cardiac output

(4) Speed of conduction of action potential

177. Bicuspid valve is present between

(1) Right and left atrium

(2) Right and left ventricle

(3) Right atrium and right ventricle

(4) Left atrium and left ventricle

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178. First heart sound is produced due to

(1) Closure of AV valves

(2) Opening of AV valves

(3) Closure of semilunar valves

(4) Opening of semilunar valves

179. In humans, during blood clotting, an enzyme complex, thrombokinase is responsible for

(1) Release of heparin from monocytes in blood

(2) Conversion of prothrombin into thrombin

(3) Conversion of thrombin into prothrombin

(4) Conversion of fibrinogen into fibrin

180. Pulmonary veins from lungs are known for carrying A blood to the B of the heart. Choose the option that fills A and B **correctly**.

	A	B
(1)	Oxygenated	Right atrium
(2)	Deoxygenated	Right ventricle
(3)	Oxygenated	Left atrium
(4)	Deoxygenated	Left ventricle

(1) (1)

(2) (2)

(3) (3)

(4) (4)

181. The hepatic portal vein carries blood from 'a' to 'b' before it is delivered to the systemic circulation.
Select the option that **correctly** identifies 'a' and 'b' respectively

(1) Intestine, kidney

(2) Kidney, intestine

(3) Liver, intestine

(4) Intestine, liver

182. Closed circulatory system with 4 chambered heart is found in all of the following, **except**

(1) Birds

(2) Mammals

(3) Amphibians

(4) Crocodiles

183. In a blood transfusion case to avoid clumping of blood in a patient having blood group A, the donor blood group should be

(1) A and AB

(2) A, AB and B

(3) A and O

(4) A, B and O

184. **Statement A:** End of P-wave represents the end of atrial diastole.
Statement B: T-wave represents ventricular repolarization.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

185. Select the **incorrect** statement w.r.t. humans.

(1) SA node generates the maximum number of action potential per minute so it sets the pace of the activities of the heart

(2) About 70 ml of blood is pumped out by each ventricle during a cardiac cycle which is called stroke volume

(3) Heart is autoexcitable, its functions can be moderated by neural and hormonal mechanisms

(4) Systemic heart includes the right ventricle and left atrium

SECTION-B

186. Category of storage polysaccharides does **not** include

(1) Inulin

(2) Chitin

(3) Glycogen

(4) Starch

187. When iodine solution is added to cellulose, the colour of reaction mixture

(1) Changes to red colour

(2) Remains unchanged

(3) Changes to blue colour

(4) Changes to yellow colour

188. How many of following are included in the category of dense connective tissue?
(a) Areolar tissue
(b) Adipose tissue
(c) Tendon
(d) Ligament
Select the correct option.

(1) One

(2) Two

(3) Three

(4) Four

189. Read the following statements and select the correct option.
Statement A: Fibres of muscular tissue exhibit excitability and contractility.
Statement B: All cells of nervous tissue exhibit excitability and conductivity.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct but statement B is incorrect

(4) Statement A is incorrect but statement B is correct

190. Read the following statements :
(i) Tight junctions stop substances from leaking across a tissue
(ii) Adhering junctions perform cementing to keep neighbouring cells together.
(iii) Gap junctions help adjacent cells to communicate for rapid transfer of macromolecules only.
(iv) Tight, adhering and gap junctions are found only in the epithelial tissue.
Choose the option that represents **incorrect** statements only

(1) (i) and (ii)

(2) (ii) and (iii)

(3) (iii) and (iv)

(4) (ii) and (iv)

191. The element present in greater concentration in human body than Earth's crust is

(1) Sodium

(2) Nitrogen

(3) Magnesium

(4) Silicon

192. Which of the following is present in retentate portion during analysis of biomolecules?

(1) Glucose

(2) Guanine

(3) Alanine

(4) Haemoglobin

193. In a healthy adult man, the partial pressure of oxygen in the systemic arteries is 95 mm Hg which is

(1) Less than the pO_2 in lung alveoli

(2) Less than the pCO_2 in systemic veins

(3) Equal to the pCO_2 in systemic veins

(4) Less than the pO_2 in actively metabolising tissues

194. Which of the following gases has 20-25 times higher solubility in human blood than that of O_2?

(1) CO_2

(2) H_2

(3) CO

(4) N_2

195. A person tries to hold his breath for one minute. Which of the following changes in blood would first lead to an urge to breathe?

(1) Fall in O_2 concentration

(2) Rise in CO_2 concentration

(3) Rise in pH

(4) Fall in osmolarity

196. Consider the reaction given below:

$$CO_2 + H_2O \overset{1}{\rightleftarrows} H_2CO_3 \overset{2}{\rightleftarrows} HCO_3^- + H^+$$

Select the **correct** option.

(1) The reaction proceeds towards the right side when the levels of the carbon-dioxide are low

(2) The enzyme catalysing the reaction 1 belongs to the VIth class of enzymes

(3) The above reaction occurs mainly in the blood plasma.

(4) The reaction proceeds in the reverse direction at the site where exchange of gases occurs in between the blood and alveolar air

197. Which of the following volume/event remains unchanged during strenuous exercise?

(1) Sequence of events in cardiac cycle

(2) Stroke volume

(3) Duration of cardiac cycle

(4) Cardiac output

198. **Assertion (A):** Fats are absorbed through lymph in the lacteals present in the intestinal villi.
Reason (R): Lymph is not a carrier for nutrients and hormones.
In the light of above statements, choose the correct option.

(1) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

199. Patients 'X' and 'Y' are hospitalised. The heart of 'X' is not pumping blood effectively enough to meet the body's need while the heart muscles in 'Y' is unexpectedly damaged by insufficient blood supply.
Identify the diseases by which 'X' and 'Y' are suffering from respectively and choose the **correct** option.

(1) Cardiac arrest, heart failure

(2) Heart failure, heart attack

(3) Heart attack, angina pectoris

(4) Angina pectoris, heart failure

200. If due to some injury the chordae tendinae of mitral valve of the heart become partially non-functional or are completely damaged, what will be the immediate effect?

(1) The flow of blood into the pulmonary artery will be reduced

(2) The flow of blood into the aorta will be reduced

(3) The pacemaker will stop functioning

(4) The blood will tend to flow back into the right atrium

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NCERT BOOSTER TEST SERIES

(for NEET-2025)

Test - 1

Answers

1. (1)	41. (3)	81. (3)	121. (2)	161. (1)
2. (1)	42. (2)	82. (1)	122. (4)	162. (4)
3. (1)	43. (2)	83. (3)	123. (2)	163. (2)
4. (2)	44. (3)	84. (2)	124. (1)	164. (2)
5. (4)	45. (2)	85. (1)	125. (3)	165. (1)
6. (1)	46. (2)	86. (2)	126. (1)	166. (4)
7. (3)	47. (3)	87. (2)	127. (3)	167. (3)
8. (3)	48. (3)	88. (3)	128. (1)	168. (2)
9. (2)	49. (2)	89. (1)	129. (2)	169. (2)
10. (3)	50. (2)	90. (2)	130. (3)	170. (3)
11. (1)	51. (1)	91. (2)	131. (3)	171. (1)
12. (3)	52. (2)	92. (4)	132. (2)	172. (1)
13. (1)	53. (2)	93. (2)	133. (3)	173. (3)
14. (2)	54. (3)	94. (1)	134. (3)	174. (2)
15. (1)	55. (2)	95. (2)	135. (1)	175. (4)
16. (2)	56. (4)	96. (4)	136. (3)	176. (2)
17. (2)	57. (1)	97. (4)	137. (3)	177. (4)
18. (3)	58. (4)	98. (3)	138. (1)	178. (1)
19. (3)	59. (1)	99. (4)	139. (2)	179. (2)
20. (2)	60. (3)	100. (2)	140. (2)	180. (3)
21. (3)	61. (2)	101. (2)	141. (1)	181. (4)
22. (3)	62. (2)	102. (1)	142. (4)	182. (3)
23. (4)	63. (4)	103. (1)	143. (3)	183. (3)
24. (1)	64. (4)	104. (3)	144. (4)	184. (4)
25. (1)	65. (4)	105. (3)	145. (3)	185. (4)
26. (2)	66. (2)	106. (2)	146. (1)	186. (2)
27. (2)	67. (3)	107. (4)	147. (3)	187. (2)
28. (4)	68. (2)	108. (1)	148. (3)	188. (2)
29. (4)	69. (3)	109. (3)	149. (2)	189. (3)
30. (2)	70. (4)	110. (2)	150. (3)	190. (3)
31. (1)	71. (1)	111. (1)	151. (1)	191. (2)
32. (2)	72. (2)	112. (2)	152. (3)	192. (4)
33. (3)	73. (3)	113. (4)	153. (3)	193. (1)
34. (1)	74. (4)	114. (4)	154. (4)	194. (1)
35. (1)	75. (1)	115. (1)	155. (3)	195. (2)
36. (3)	76. (3)	116. (1)	156. (4)	196. (4)
37. (1)	77. (4)	117. (4)	157. (4)	197. (1)
38. (3)	78. (1)	118. (4)	158. (2)	198. (3)
39. (1)	79. (3)	119. (4)	159. (1)	199. (2)
40. (2)	80. (4)	120. (1)	160. (2)	200. (2)